

Simple Linear Regression

Simple Linear Regression

- Our objective is to study the relationship between two variables X and Y .
- One way is by means of **regression**.
- Regression analysis is the process of estimating a functional relationship between X and Y . A regression equation is often used to predict a value of Y for a given value of X .
- Another way to study relationship between two variables is **correlation**. It involves measuring the direction and the strength of the **linear** relationship.

Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where

y = dependent variable

x = independent variable

β_0 = y-intercept

β_1 = slope of the line

ε = error variable

Simple Linear Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

This model is

- **Simple:** only one X
- **Linear in the parameters:** No parameter appears as exponent or is multiplied or divided by another parameter
- **Linear in the predictor variable (X):** X appears only in the first power.

Examples

- Multiple Linear Regression:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

- Polynomial Linear Regression:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

- Linear Regression:

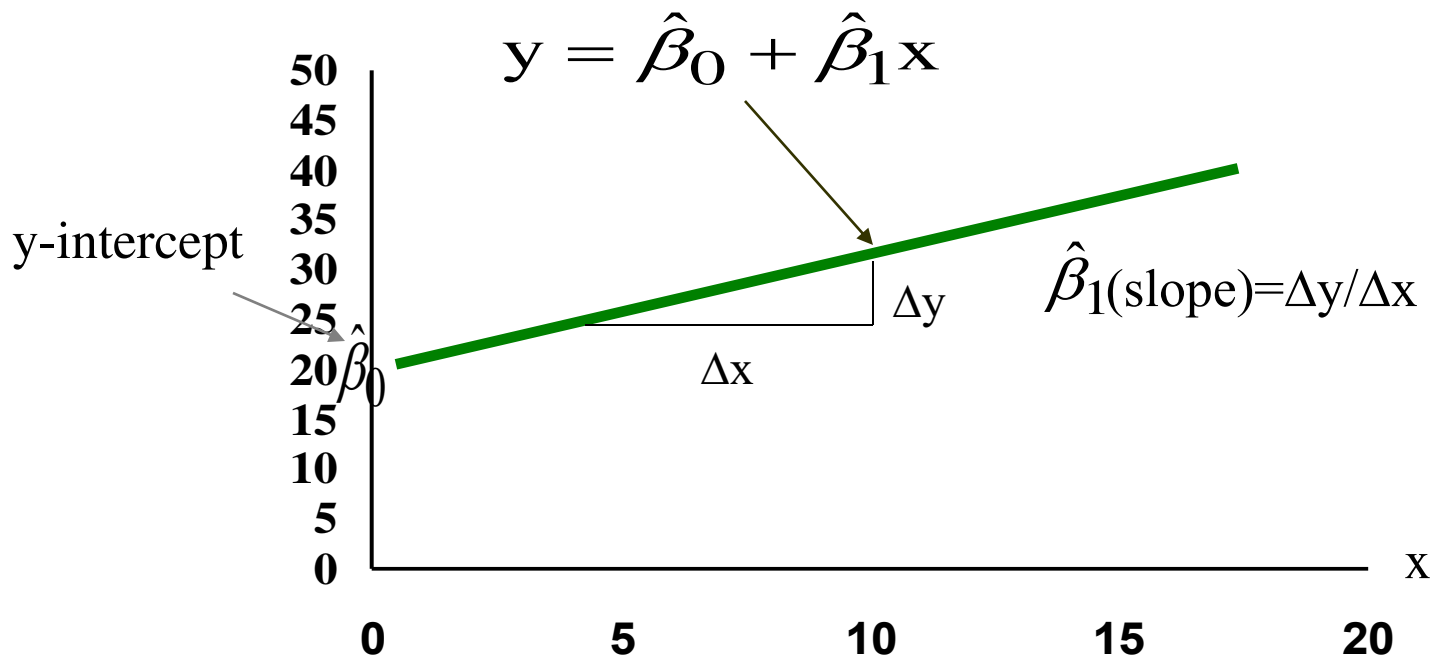
$$\log_{10}(Y_i) = \beta_0 + \beta_1 X_i + \beta_2 \exp(X_i) + \varepsilon_i$$

- Nonlinear Regression:

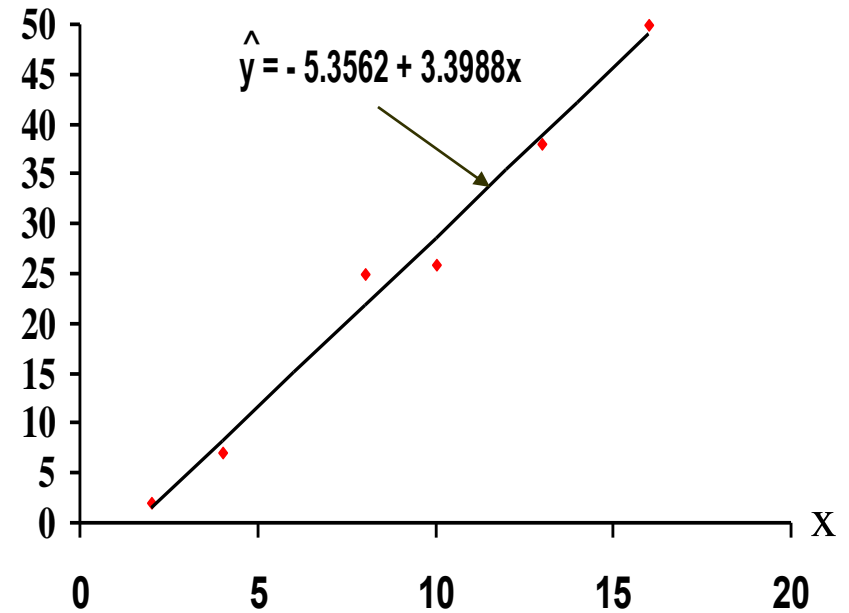
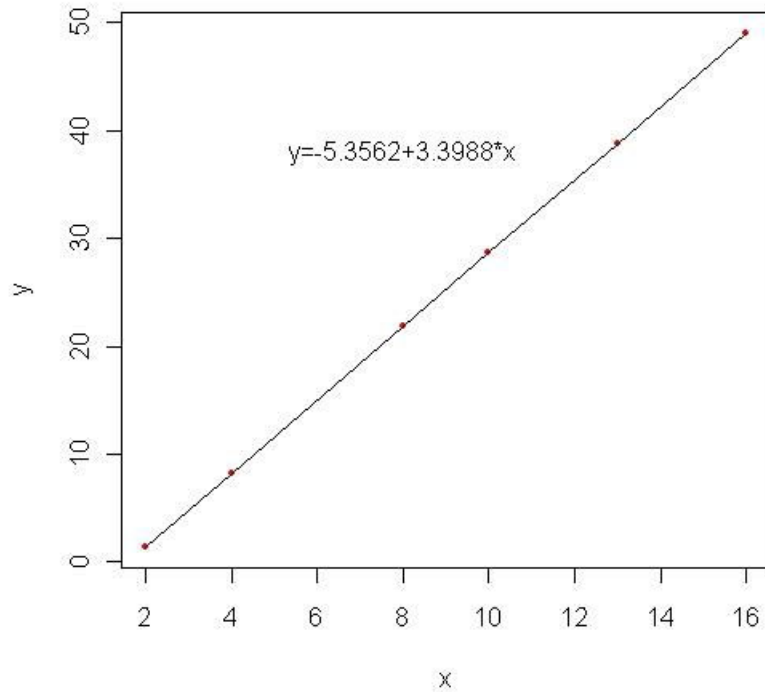
$$Y_i = \beta_0 / (1 + \beta_1 \exp(\beta_2 X_i)) + \varepsilon_i$$

Linear or nonlinear in parameters

Deterministic Component of Model



Mathematical vs Statistical Relation



Error

- **The scatterplot shows that the points are not on a line, and so, in addition to the relationship, we also describe the error:**

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \text{ } i=1,2,\dots,n$$

- The Y's are the response (or dependent) variable. The x's are the predictors or independent variables, and the epsilon's are the errors. We assume that the errors are normal, mutually independent, and have variance σ^2 .

Least Squares:

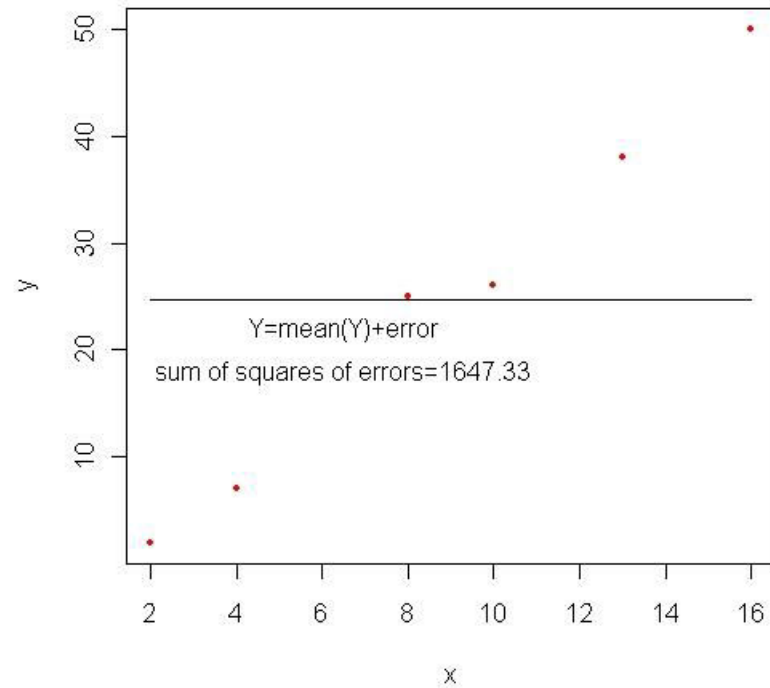
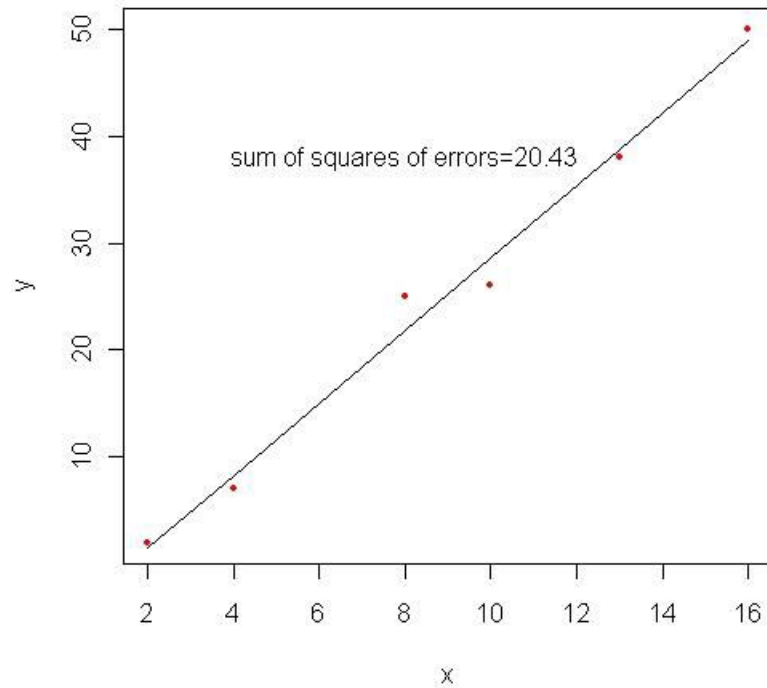
Minimize
$$\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\Rightarrow \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- The quantities $R_i = y_i - \hat{y}_i$ are called the residuals. If we assume a normal error, they should look normal.

Error: $Y_i - E(Y_i)$ **unknown**; Residual: $R_i = y_i - \hat{y}_i$ estimated, i.e. **known**

Minimizing error



- The Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- The Least Squares Regression Line where

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$SS_x = \sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

$$SS_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

What form does the error take?

- Each observation may be decomposed into two parts:

$$y = \hat{y} + (y - \hat{y})$$

- The first part is used to determine the fit, and the second to estimate the error.
- We estimate the standard deviation of the error by:

$$SSE = \sum (Y - \hat{Y})^2 = S_{yy} - \left(\frac{S_{xy}^2}{S_{xx}} \right)$$

Estimate of σ^2

- We estimate σ^2 by

$$s^2 = \frac{SSE}{n-2} = MSE$$

Example

- An educational economist wants to establish the relationship between an individual's income and education. He takes a random sample of 10 individuals and asks for their income (in \$1000s) and education (in years). The results are shown below. Find the least squares regression line.

Education	11	12	11	15	8	10	11	12	17	11
Income	25	33	22	41	18	28	32	24	53	26

Dependent and Independent Variables

- The dependent variable is the one that we want to forecast or analyze.
- The independent variable is hypothesized to affect the dependent variable.
- In this example, we wish to analyze income and we choose the variable individual's education that most affects income. Hence, y is income and x is individual's education

First Step:

$$\sum x_i = 118$$

$$\sum x_i^2 = 1450$$

$$\sum y_i = 302$$

$$\sum y_i^2 = 10072$$

$$\sum x_i y_i = 3779$$

Sum of Squares:

$$SS_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 3779 - \frac{(118)(302)}{10} = 215.4$$

$$SS_x = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 1450 - \frac{(118)^2}{10} = 57.6$$

Therefore, $\hat{\beta}_1 = \frac{SS_{xy}}{SS_x} = \frac{215.4}{57.6} = 3.74$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{302}{10} - 3.74 \frac{118}{10} = -13.93$$

The Least Squares Regression Line

- The least squares regression line is

$$\hat{y} = -13.93 + 3.74x$$

- Interpretation of coefficients:

- *The sample slope $\hat{\beta}_1 = 3.74$ tells us that on average for each additional year of education, an individual's income rises by \$3.74 thousand.
- The y-intercept is $\hat{\beta}_0 = -13.93$. This value is the expected (or average) income for an individual who has 0 education level (which is meaningless here)

Example

- Car dealers across North America use the red book to determine a car's selling price on the basis of important features. One of these is the car's current odometer reading.
- To examine this issue 100 three-year old cars in mint condition were randomly selected. Their selling price and odometer reading were observed.

Portion of the data file

Odometer	Price
37388	5318
44758	5061
45833	5008
30862	5795
.....	...
34212	5283
33190	5259
39196	5356
36392	5133

Example (Minitab Output)

Regression Analysis

The regression equation is

$$\text{Price} = 6533 - 0.0312 \text{ Odometer}$$

Predictor	Coef	StDev	T	P
Constant	6533.38	84.51	77.31	0.000 (SIGNIFICANT)
Odometer	-0.031158	0.002309	-13.49	0.000 (SIGNIFICANT)

S = 151.6 R-Sq = 65.0% R-Sq(adj) = 64.7%

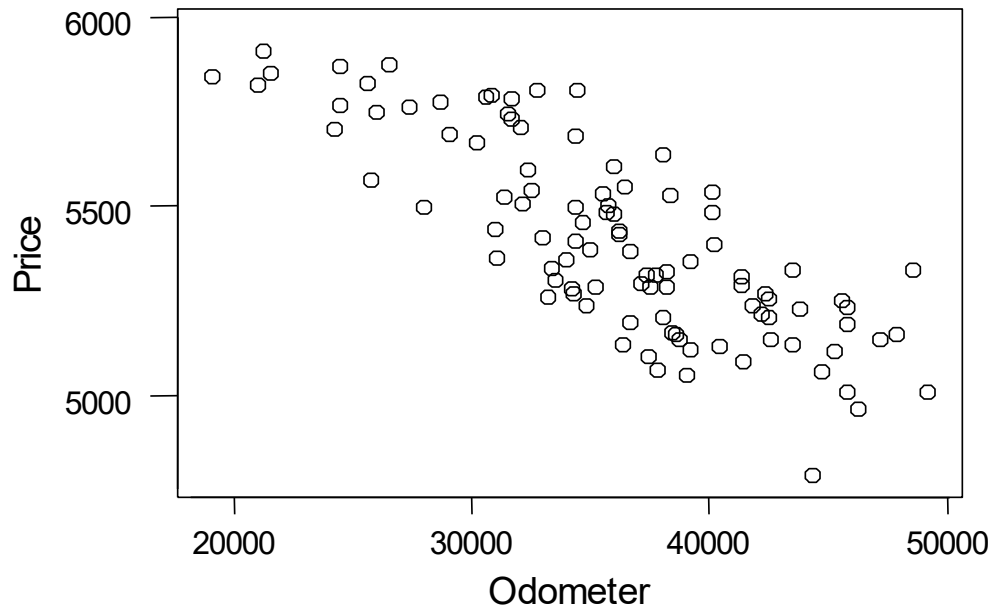
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	4183528	4183528	182.11	0.000
Error	98	2251362	22973		
Total	99	6434890			

Example

- The least squares regression line is

$$\hat{y} = 6533.38 - 0.031158x$$



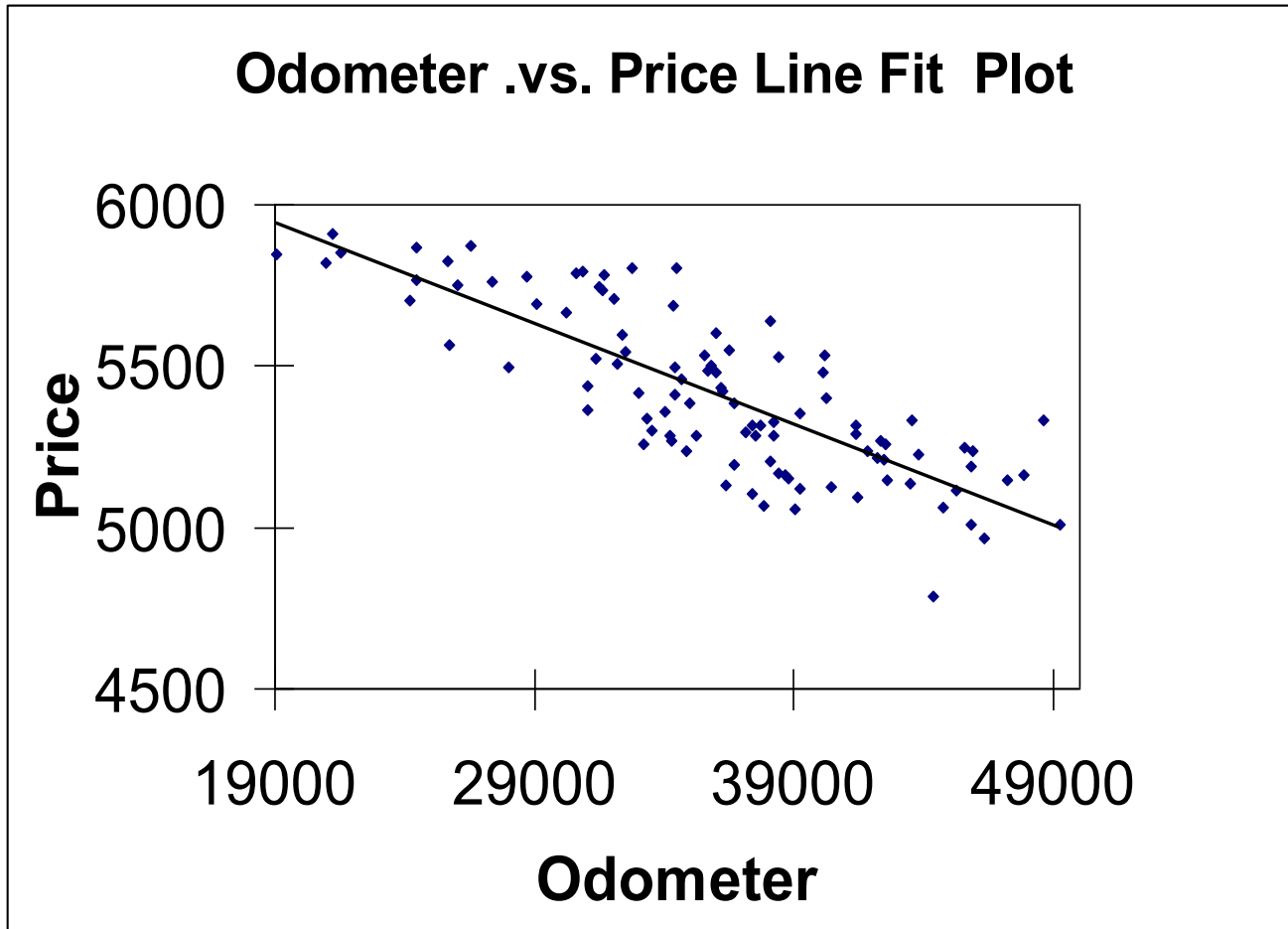
Interpretation of the coefficients

- $\hat{\beta}_1 = -0.031158$ means that for each additional mile on the odometer, the price decreases by an average of 3.1158 cents.
- $\hat{\beta}_0 = 6533.38$ means that when $x = 0$ (new car), the selling price is \$6533.38 but $x = 0$ is not in the range of x . So, we cannot interpret the value of y when $x=0$ for this problem.
- $R^2=65.0\%$ means that 65% of the variation of y can be explained by x . The higher the value of R^2 , the better the model fits the data.

R^2 and R^2 adjusted

- R^2 measures the degree of linear association between X and Y .
- So, an R^2 close to 0 does not necessarily indicate that X and Y are unrelated (relation can be nonlinear)
- Also, a high R^2 does not necessarily indicate that the estimated regression line is a good fit.
- As more and more X 's are added to the model, R^2 always increases. R^2_{adj} accounts for the number of parameters in the model.

Scatter Plot



Testing the slope

- Are X and Y linearly related?

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

• **Test Statistic:**

$$t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} \quad \text{where} \quad s_{\hat{\beta}_1} = \frac{s_{\varepsilon}}{\sqrt{SS_x}}$$

Testing the slope (continue)

- The Rejection Region:

Reject H_0 if $t < -t_{\alpha/2, n-2}$ or $t > t_{\alpha/2, n-2}$.

- If we are testing that high x values lead to high y values, $H_A: \beta_1 > 0$. Then, the rejection region is $t > t_{\alpha, n-2}$.
- If we are testing that high x values lead to low y values or low x values lead to high y values, $H_A: \beta_1 < 0$. Then, the rejection region is $t < -t_{\alpha, n-2}$.

Assessing the model

Example:

- Excel output

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	6533.4	84.512322	77.307	1E-89
Odometer	-0.031	0.0023089	-13.49	4E-24

- Minitab output

Predictor	Coef	StDev	T	P
Constant	6533.38	84.51	77.31	0.000
Odometer	-0.031158	0.002309	-13.49	0.000

Coefficient of Determination

$$R^2 = 1 - \frac{SSE}{SS_y}$$

For the data in odometer example, we obtain:

$$\begin{aligned} R^2 &= 1 - \frac{SSE}{SS_y} = 1 - \frac{2,251,363}{6,434,890} \\ &= 1 - 0.3499 = 0.6501 \end{aligned}$$

$$R^2_{\text{adj}} = 1 - \left(\frac{n-1}{n-p} \right) \frac{SSE}{SS_y}$$

where p is number of predictors in the model.

Using the Regression Equation

- Suppose we would like to predict the selling price for a car with 40,000 miles on the odometer

$$\begin{aligned}\hat{y} &= 6,533 - 0.0312x \\ &= 6,533 - 0.0312(40,000) \\ &= \$5,285\end{aligned}$$

Prediction and Confidence Intervals

- **Prediction Interval of y for $x=x_g$:** The confidence interval for *predicting the particular value of y* for a given x

$$\hat{y} \pm t_{\alpha/2, n-2} s_e \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}}$$

- **Confidence Interval of $E(y|x=x_g)$:** The confidence interval for *estimating the expected value of y* for a given x

$$\hat{y} \pm t_{\alpha/2, n-2} s_e \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}}$$

Solving by Hand (Prediction Interval)

- From previous calculations we have the following estimates:

$$\hat{y} = 5285, s_{\varepsilon} = 151.6, SS_x = 4309340160, \bar{x} = 36,009$$

- Thus a 95% **prediction interval** for $x=40,000$ is:

$$5,285 \pm 1.984(151.6) \sqrt{1 + \frac{1}{100} + \frac{(40,000 - 36,009)^2}{4,309,340,160}}$$

$$5,285 \pm 303$$

- The prediction is that the selling price of the car will fall between \$4982 and \$5588.

Solving by Hand (Confidence Interval)

- A 95% confidence interval of

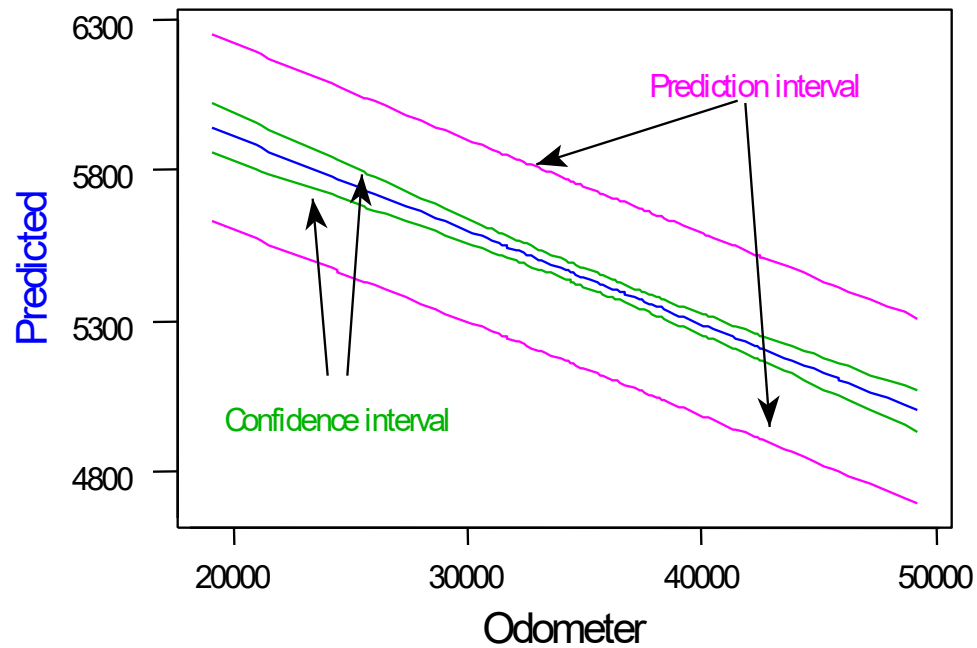
$E(y | x=40,000)$ is:

$$5,285 \pm 1.984(151.6) \sqrt{\frac{1}{100} + \frac{(40,000 - 36,009)^2}{4,309,340,160}}$$

$$5,285 \pm 35$$

- The mean selling price of the car will fall between \$5250 and \$5320.

Prediction and Confidence Intervals' Graph



Notes

- No matter how strong is the statistical relation between X and Y , no cause-and-effect pattern is necessarily implied by the regression model. Ex: Although a positive and significant relationship is observed between vocabulary (X) and writing speed (Y), this does *not* imply that an increase in X *causes* an increase in Y . Other variables, such as age, may affect both X and Y . Older children have a larger vocabulary and faster writing speed.

Regression Diagnostics

Residual Analysis:

⇒ **Non-normality**

⇒ **Heteroscedasticity (non-constant variance)**

⇒ **Non-independence of the errors**

⇒ **Outlier**

⇒ **Influential observations**

Standardized Residuals

- The standardized residuals are calculated as

$$\text{Standardized residual} = \frac{r_i}{s_\varepsilon}$$

where $r_i = y_i - \hat{y}_i$.

- The standard deviation of the i-th residual is

$$s_{r_i} = s_\varepsilon \sqrt{1 - h_i} \quad \text{where} \quad h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{SS_x}$$

Non-normality:

- The errors should be normally distributed. To check the normality of errors, we use **histogram of the residuals** or **normal probability plot of residuals** or **tests such as Shapiro-Wilk test**.
- Dealing with non-normality:
 - Transformation on Y
 - Other types of regression (e.g., Poisson or Logistic ...)
 - Nonparametric methods (e.g., nonparametric regression(i.e. smoothing))

Non-constant variance:

- The error variance σ_{ε}^2 should be constant.
- To diagnose non-constant variance, one method is to **plot the residuals against the predicted value of y (or x)**. If the points are distributed evenly around the expected value of errors which is 0, this means that the error variance is constant. Or, formal tests such as: Breusch-Pagan test

Dealing with non-constant variance

- Transform Y
- Re-specify the Model (e.g., Missing important X's?)
- Use Weighted Least Squares instead of Ordinary Least Squares

$$\min \sum_{i=1}^n \frac{\varepsilon_i^2}{\text{Var}(\varepsilon_i)}$$

Non-independence of error variable:

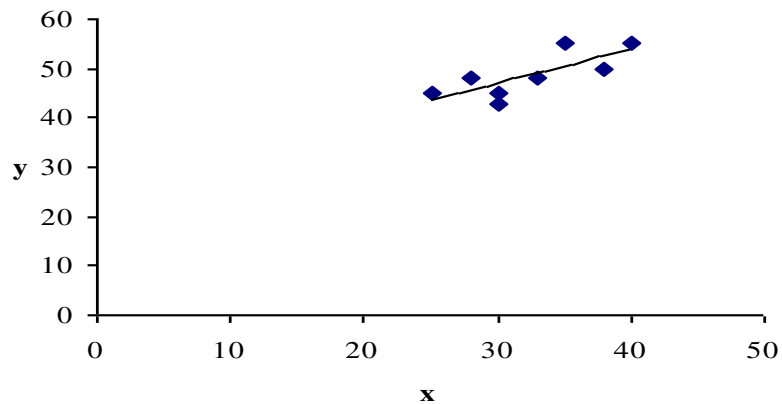
- The values of error should be independent. When the data are time series, the errors often are correlated (i.e., autocorrelated or serially correlated). To detect autocorrelation we plot the residuals against the time periods. If there is no pattern, this means that errors are independent. Or, more formal tests such as Durbin-Watson

Outlier:

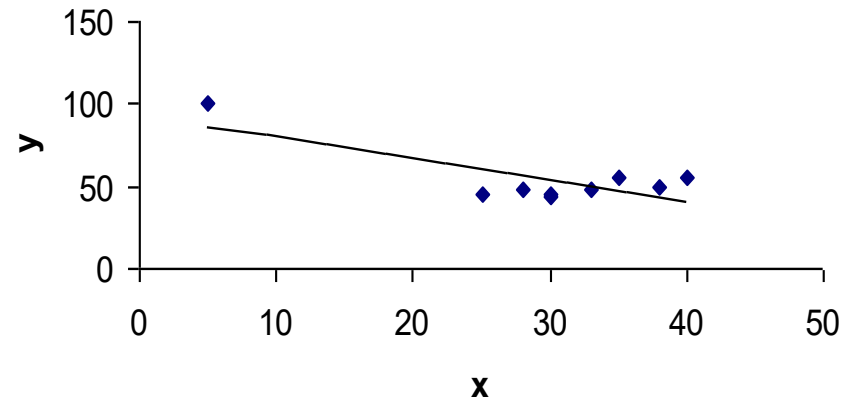
- An outlier is an observation that is unusually small or large. Two possibilities which cause outlier is
 1. Error in recording the data. $\Rightarrow$ Detect the error and correct it
The outlier point should not have been included in the data (belongs to another population) $\Rightarrow$ Discard the point from the sample
 2. The observation is unusually small or large although it belong to the sample and there is no recording error. $\Rightarrow$ Do NOT remove it

Influential Observations

Scatter Plot Without the Influential Observation



Scatter Plot of One Influential Observation



Influential Observations

- Detection:

Cook's Distance, DFFITS, DFBETAS (Neter, J.,
Kutner, M.H., Nachtsheim, C.J., and Wasserman, W., (1996)
Applied Linear Statistical Models, 4th edition, Irwin, pp. 378-384)

Multicollinearity

- A common issue in multiple regression is multicollinearity. This **exists when some or all of the predictors in the model are highly correlated**. In such cases, the estimated coefficient of any variable depends on which other variables are in the model. Also, standard errors of the coefficients are very high...

Multicollinearity

- Look into correlation coefficient among X's: If $\text{Cor} > 0.8$, suspect multicollinearity
- Look into Variance inflation factors (VIF): $\text{VIF} > 10$ is usually a sign of multicollinearity
- If there is multicollinearity:
 - Use transformation on X's, e.g. centering, standardization. Ex: $\text{Cor}(X, X^2) = 0.991$; after standardization $\text{Cor} = 0$!
 - Remove the X that causes multicollinearity
 - Factor analysis
 - Ridge regression
 - ...

Exercise

- In baseball, the fans are always interested in determining which factors lead to successful teams. The table below lists the team batting average and the team winning percentage for the 14 league teams at the end of a recent season.

Team-B-A	Winning%
0.254	0.414
0.269	0.519
0.255	0.500
0.262	0.537
0.254	0.352
0.247	0.519
0.264	0.506
0.271	0.512
0.280	0.586
0.256	0.438
0.248	0.519
0.255	0.512
0.270	0.525
0.257	0.562

y = winning % and x = team batting average

a) LS Regression Line

$$\sum x_i = 3.642, \sum x_i^2 = 0.949$$

$$\sum y_i = 7.001, \sum y_i^2 = 3.549$$

$$\sum x_i y_i = 1.824562$$

$$SS_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 1.824562 - \frac{(3.642)(7.001)}{14} = 0.0033$$

$$SS_x = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 0.948622 - \frac{(3.642)^2}{14} = 0.00118$$

$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_x} = \frac{0.003302}{0.001182} = 0.7941$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.5 - (0.7941)0.26 = 0.2935$$

- The least squares regression line is

$$\hat{y} = 0.2935 + 0.7941x$$

- The meaning $\hat{\beta}_1 = 0.7941$ is for each additional batting average of the team, the winning percentage increases by an average of 79.41%.

b) Standard Error of Estimate

$$\begin{aligned}SSE &= S_{yy} - \left(\frac{S_{xy}^2}{S_{xx}} \right) = \left(\sum y_i^2 - \frac{(\sum y_i)^2}{n} \right) - \left(\frac{S_{xy}^2}{S_{xx}} \right) \\&= (3.548785 - \frac{7.001^2}{14}) - \frac{0.003302^2}{0.00182} = 0.03856\end{aligned}$$

$$\text{So, } s_{\varepsilon}^2 = \frac{SSE}{n-2} = \frac{0.03856}{14-2} = 0.00321 \text{ and } s_{\varepsilon} = \sqrt{s_{\varepsilon}^2} = 0.0567$$

- Since $s_{\varepsilon}=0.0567$ is small, we would conclude that “s” is relatively small, indicating that the regression line fits the data quite well.

c) Do the data provide sufficient evidence at the 5% significance level to conclude that higher team batting average lead to higher winning percentage?

$$\begin{array}{ll} H_0 : & \beta_1 = 0 \\ H_A : & \beta_1 > 0 \end{array} \quad \text{Test statistic: } t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} = 1.69 \quad (\text{p-value}=.058)$$

Conclusion: Do not reject H_0 at $\alpha = 0.05$.
The higher team batting average does not lead to higher winning percentage.

d) Coefficient of Determination

$$R^2 = \frac{SS_{xy}^2}{SS_x - SS_y} = 1 - \frac{SSE}{SS_y} = 1 - \frac{0.03856}{0.04778} = 0.1925$$

The 19.25% of the variation in the winning percentage can be explained by the batting average.

e) Predict with 90% confidence the winning percentage of a team whose batting average is 0.275.

$$\hat{y} = 0.2935 + 0.7941(0.275) = 0.5119$$

$$\hat{y} \pm t_{\alpha/2, n-2} s_{\varepsilon} \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{SS_x}} =$$

$$0.5119 \pm (1.782)(0.0567) \sqrt{1 + \frac{1}{14} + \frac{(0.275 - 0.2601)^2}{0.001182}}$$

$$0.5119 \pm 0.1134$$

90% PI for y: (0.3985, 0.6253)

- The prediction is that the winning percentage of the team will fall between 39.85% and 62.53%.